

GE Healthcare Coils

GE Signa[®] 1.5T 8-Channel Cardiac Array Service Manual

GE Catalog Part Number: M3335JD



2304255-5
Rev 6
© 2009, General Electric Company,
All Rights Reserved

Revision History

Date	Rev	ECO	Change Description
10/03	1	N/A	First Issue
10/07	2	N/A	Added, "Please see the locations of the preamps shown in Figure 9A,B below. The wrist strap MUST be worn all times when handling Preamps" on page 16. Added figure 9A & B to page 17. Re-numbered figures.
11/07	3	2007460	Applied New Template, Added preamplifier to table 6.1 and revised Manual, Replaced the word "coil" to "image" on section 1-5 /line 2, Added the line "Note: The wrist strap MUST be worn all times when handling Coils" to sections 3-3, 5-1 and 5-2, Removed 2 channels from the table on section 3-5
08/08	4	2008215	Table 1-1: Revised p/n's listed in table. Table 6-1: Removed Preamplifier. Table 6-2: Added GE p/n for Pad Kit. Section 1-3: Changed Op manual p/n. Section 3-2-3: Revised step 5. Section 3-5: Revised Symptom #2 table. Removed Figure 9A nd 9B.
05/09	5	23717797	Replace U1-150478 with 2423518, U1-150497 with 2423519 and U1-150477 with 2423523 in Shipping List – Table 1-1. Add Row for Patient comfort pad U1-155164 and remove U1-155164 from pad kit row on Table 6-2. Add Croatian Service Disclaimer. Updated Figure 1.
11/09	6		Updated damage in transportation, language policy, shipping contents, and new MCQA procedure for Unified phantom set per the Phantom Elimination Program.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. 6 November 12, 2009

LANGUAGE POLICY FOR SERVICE DOCUMENTATION (DOC0620154 REV 1)

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatt sérülését eredményezheti.
ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. - Reynið ekki að afgangið tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION	12
1-1 Product Identification and Shipping List	12
Note: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR System.	13
1-2 Compatibility	13
1-3 Related Documentation	13
1-4 Environmental Requirements	13
1-5 Theory of Operation	14
SECTION 2 – SETUP AND CALIBRATION	17
2-1 Coil Installation	17
2-1-1 Special Install Notes	17
2-1-2 Configuration for 11x Software	17
2-1-3 Configuration for 10x Software	17
2-2 Installation Functional Checks	17
2-3 Periodic Quality Assurance Check.....	18
SECTION 3 – FUNCTIONAL CHECKS	19
3-1 Scanner Verification	19
3-2 Coil Imaging Performance Verification	19
3-2-1 Tools Required.....	19
3-2-2 Explanation of Procedure	19
3-2-3 SNR Setup for 11.0 M3A and Greater Software Versions with Legacy Phantom	20
3-2-4 SNR Setup for 11.0 M3A and Greater Software Versions with Unified Phantom Set	21
3-2-5 MCQA / SNR Test.....	23
3-3 External Cable Check	23
3-4 PIN Diode Check.....	24
3-6 Coil Element Information	26
SECTION 4 – MAINTENANCE	27
4-1 Coil Care	27
4-2 Special Care Requirements.....	27
SECTION 5 –REPLACEMENT	28
5-1 Anterior Cable Replacement	28
5-2 Posterior Cable Replacement	29
SECTION 6 – RENEWAL PARTS.....	31

6-1 Field Replaceable Units.....31

6-2 Other Replaceable Accessories.....31

7-1 SNR Data Sheet.....32

7-2 Schematic.....33

SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 8-Channel Cardiac Array, refer to the coil label (see *Figure 1*).

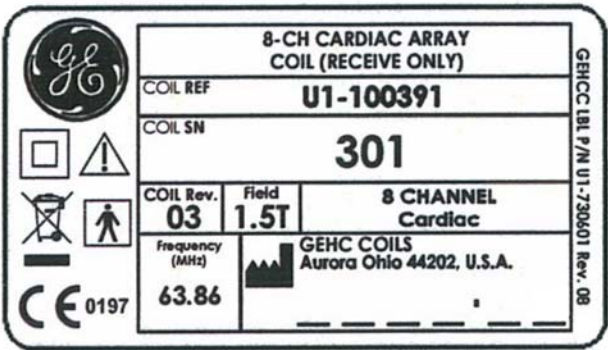


Figure 1: Coil label.

Figure 2 shows a picture of the 8-Channel Cardiac Array.



Figure 2: The 8-Channel Cardiac Array.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	N/A	U1-100391	1
2	Patient Comfort Pad	2423518	N/A	1
2	Patient Headrest Pad	2423519	N/A	1
2	Patient Lower Body Pad	2423523	N/A	1
1	Operator Guide Packet	2422147-500	N/A	1
1	Service Manual	2304255-5	N/A	1

Note: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR System.

1-2 Compatibility

The 8-Channel Cardiac Array is compatible with the Excite 1.5T system and requires 10.0 M4 or greater software. 11.0 M4 software is recommended.

The automated QA Tool is only available for software 11.0 M3 and greater.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422147.

Signa Excite Service Methods CD, GE Part Number 2333500-3.

1-4 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

Dimensions

19.4" L x 16.5" W x 4.7" H (49.4cm x 42.0cm x 11.9cm)

Weight

12 lbs (5.4 kg)

1-5 Theory of Operation

The block diagram of 8-Channel Cardiac Array is shown in *Figure 3*. The coil is designed to give a high-resolution image for the cardiovascular system. It works at 1.5T field strength and is capable of doing Array Spatial Sensitivity Encoding Technique (ASSET). This design reduces the breath-hold time, which will aid the patient to successfully hold their breath for the designated time while scanning. The coil is composed of two separate pieces, anterior and posterior. The posterior piece is rigid. The anterior piece is flexible. The flexible design of the coil can accommodate different patient habits. Thus the optimum Signal to Noise Ratio (SNR) can be achieved.

The 8-Channel Cardiac Array is a receive-only coil, as indicated in the schematic shown in the Appendix. The coil is actively decoupled from the RF transmit coil during transmit by means of a RF choking circuit. The choking circuit elements are switched on and off by pin diodes or by small-signal diodes. During body coil transmission, the pin diodes are turned on, as shorted circuits, with a forward bias DC source provided by the MR system (active decoupling). The small-signal diodes are turned on by the induced RF voltage in the chokes, coupled from the transmit field (passive decoupling). When either the pin diodes or the small-signal diodes are turned on, the RF choking elements become very high in impedance (typically above 2 kilo-ohm), compared with the other circuit elements (less than 70 ohm). These high impedance elements in the coil segregate the coil circuitry into several isolated electrical segments, preventing any current flow in the coil circuit.

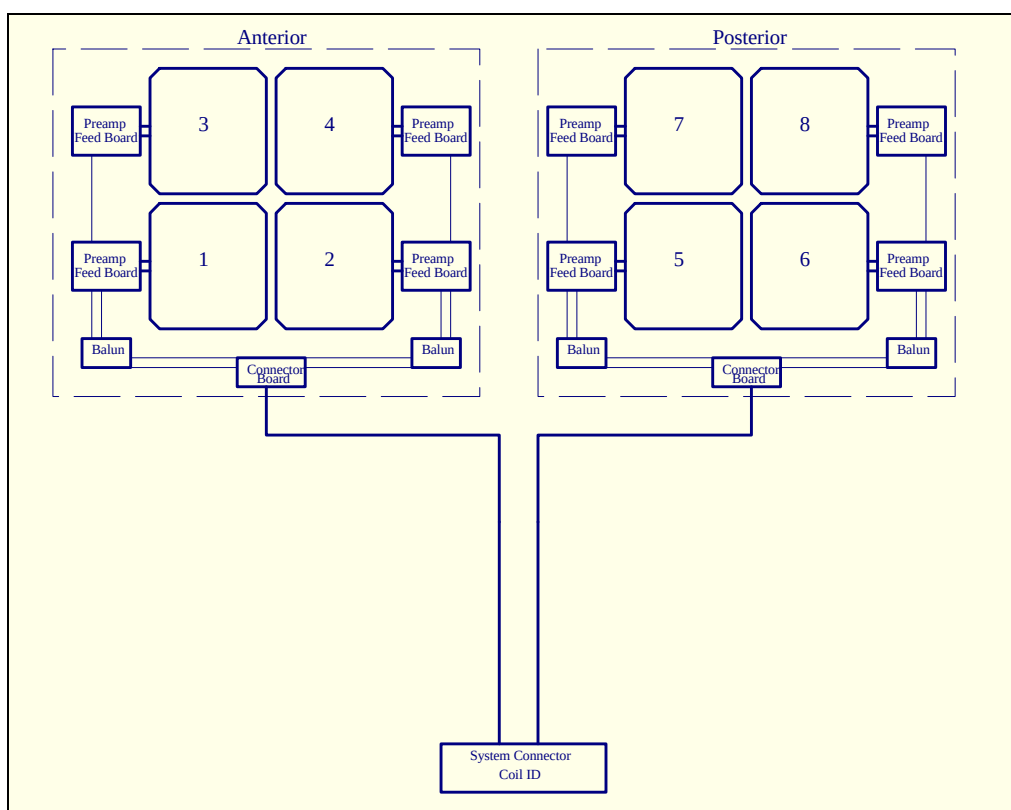


Figure 3: Block diagram of the coil.

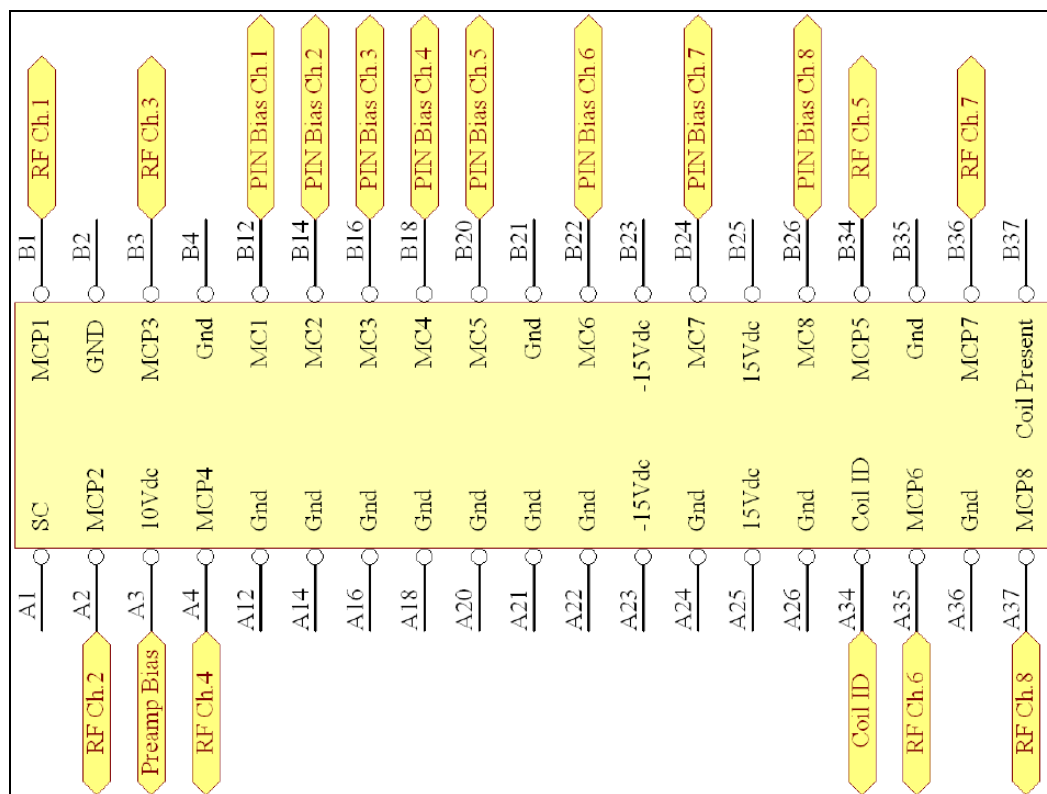


Figure 4: Block Diagram of the coil to connector interface.

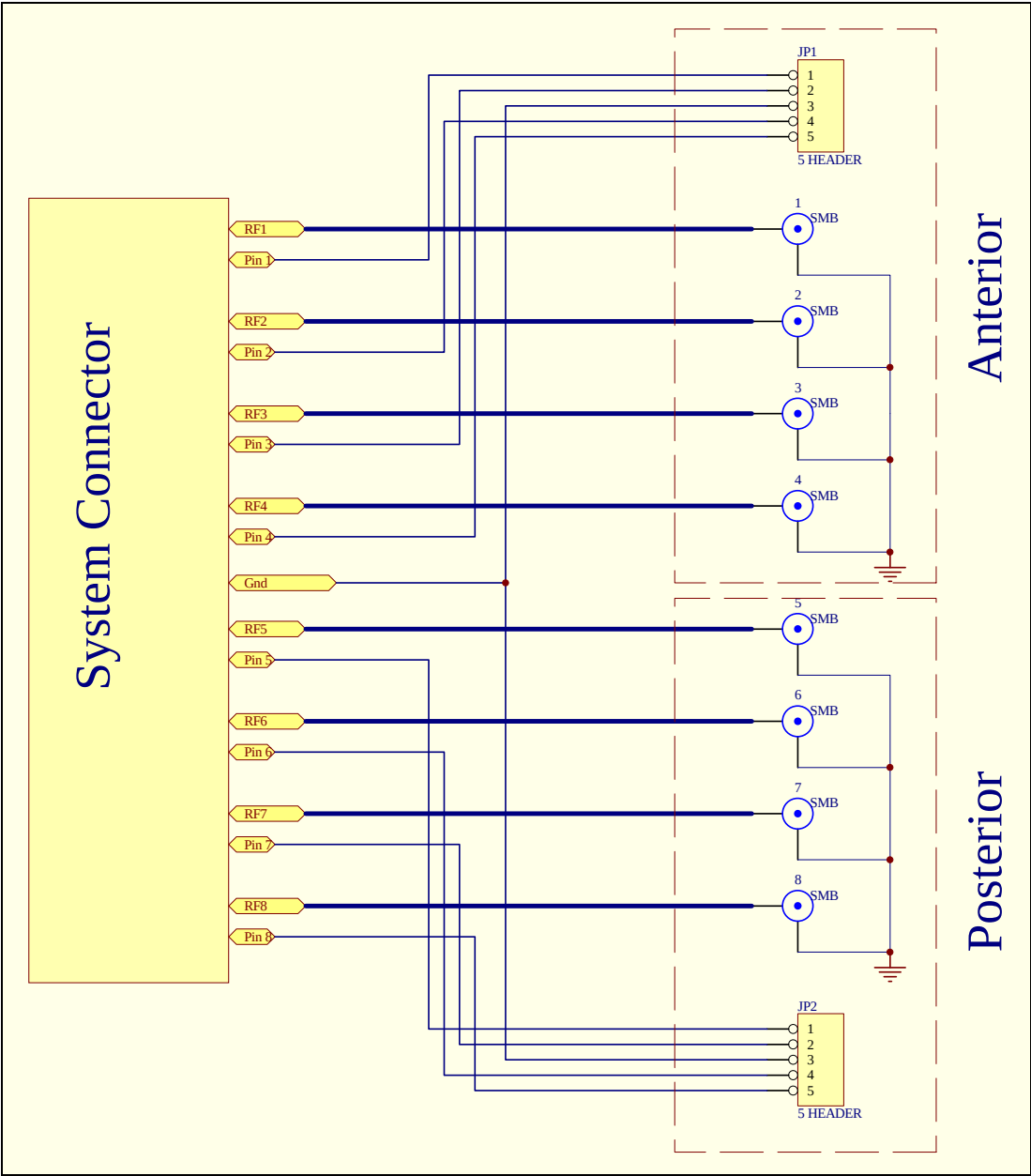


Figure 5: Block diagram of the circuit layout.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration for 11x Software

Use the Configuration File Manager to install the coil configurations listed in Table 2-1-3.

2-1-3 Configuration for 10x Software

The 8-Channel Cardiac Array requires 10.0 M4 or greater software. 11.0 M4 software is recommended.

Updated coil configurations in the system are required for software versions prior to 11.0 M4. Updated coil configurations are recommended for 11.0 M4 and greater software revisions.

To update the coil configuration files:

1. Using the configuration manager, remove all currently installed (if any) configurations for the 8-Channel Cardiac Array.
2. Update the coil configurations to the latest version (Refer to Section 9 of “Host Software Install Release 10.X”, direction OW2SCA8G. as found on the Service Methods CD, GE Part Number 2333500-3).
3. Install the new configurations listed in table 2-1-3.

CONFIGURATION MODES – TABLE 2-1-3

8CARDIAC	8CARDIACANT	8CARDIACPOST
----------	-------------	--------------

2-2 Installation Functional Checks

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Plug the coil into the patient table.

3. At the console, verify the Coil has been properly identified by the system: correct name in **Coil** field and correct picture on the screen.
4. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet (See section 7).

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM – TABLE 3-2-1-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2304255-3	U1-150511	1

TOOLS REQUIRED USING UNIFIED PHANTOM – TABLE 3-2-1-2

Description	GE Part #	Qty
TL Unified Phantom	5343347	1

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup for 11.0 M3A and Greater Software Versions with Legacy Phantom

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position the phantom set on the coil without the pad, lining up the pegs on the phantom to the holes in the posterior section of the coil (see *Figure 6*).



Figure 6: Positioning the phantom on the coil.

3. Using the raised notches on top of the phantom, align the anterior section on the phantom and fasten the Velcro straps.
4. Landmark the coil at the indicated line and advance to scan (see *Figure 7*).

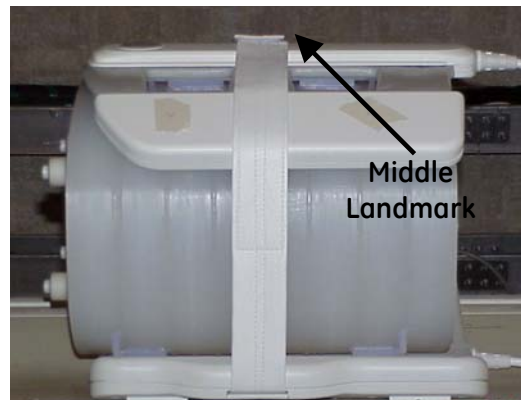


Figure 7: Landmarking the coil.

3-2-4 SNR Setup for 11.0 M3A and Greater Software Versions with Unified Phantom Set

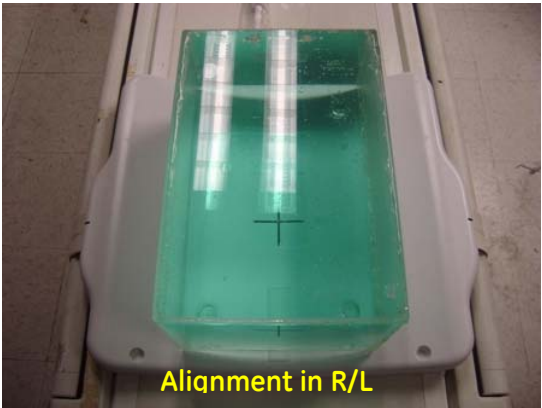
Visual	Description
 Figure 3.2.4-1	1. Place the posterior section of the coil on the table as shown in Figure 3.2.4-1.
 Figure 3.2.4-2  Figure 3.2.4-3	2. Place the TL unified phantoms over the posterior section in such a way that the phantom set is well centered in both S/I and R/L directions as shown in Figure 3.2.4-2 and Figure 3.2.4-3.



Figure 3.2.4-4

3. Place the anterior section on the phantom set such that the coil is well centered in both S/I and R/L directions with respect to the phantom set as shown in the Figure 3.2.4-4



Figure 3.2.4-5

4. Fasten the Velcro straps so that the coil firmly snugs the phantom as shown in Figure 3.2.4-5.
5. Connect the coil to respective port of the system LPCA.

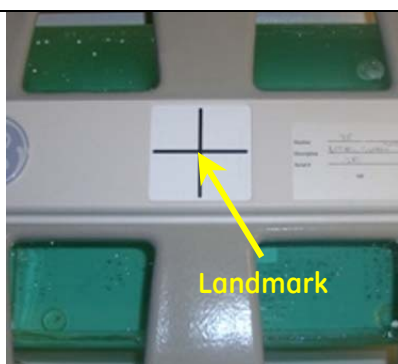


Figure 3.2.4-6

6. Landmark the coil at the cross mark as shown in Figure 3.2.4-6 and advance to scan.

3-2-5 MCQA / SNR Test

1. Move back to the operator workspace and end the previous exam, if required.
2. For 12.x or earlier systems: From the Common Service Desktop, select **Image Quality**, and then **Coil SNR**. Then select **Click here to start this tool**.

For 14.x or later systems: From the Common Service Desktop, select **Image Quality**, and then **MCQA Tool**. Then select **Click here to start this tool**.

3. Select the coil type you want to run, enter the serial number and select **Start Scan**.

Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.

3-3 External Cable Check

Diode Check from the Connector:

Note: The wrist strap **MUST** be worn all times when handling Coils.

Step 1 – Using a multi-meter set to the diode check function, check the coil diodes in the following manner.

Step 2 – Orient the connector as shown in *Figure 8*.

Step 3 – Follow Table 3-3 for the correct pins and lead locations to verify the function of the diodes.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3

PIN Number	Positive Lead	Negative Lead	DMM Reading
1	B12	A12	0.5 +/- 0.2V
2	B14	A14	0.5 +/- 0.2V
3	B16	A16	0.5 +/- 0.2V
4	B18	A18	0.5 +/- 0.2V
5	B20	A20	0.5 +/- 0.2V
6	B22	A22	0.5 +/- 0.2V
7	B24	A24	0.5 +/- 0.2V
8	B26	A26	0.5 +/- 0.2V

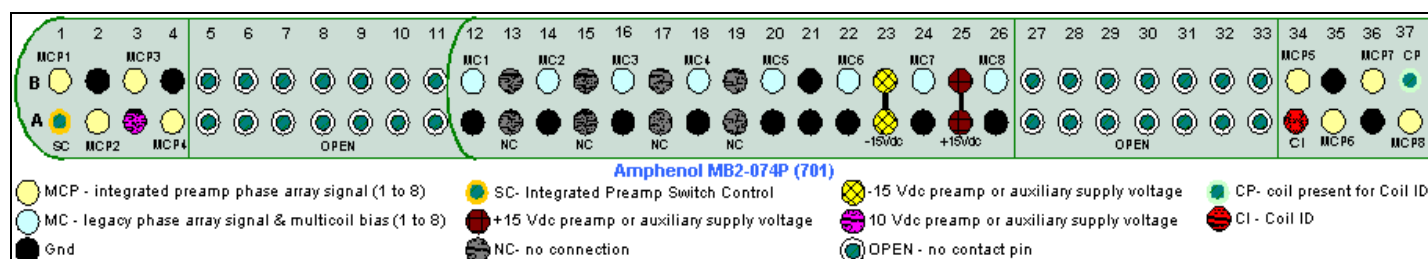


Figure 8: 60 pin Bendix connector.

Note: This connector is system side. Coil side has rows A and B reversed.

3-4 PIN Diode Check

Same as External Cable Check.



The posterior section of the coil must be plugged into the system connector for the pin diode check.

3-5 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, check the output cable by itself (see below). If the cable and PIN diodes are okay, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar coil to see if the same problem occurs.	Correct MRI system problem.
There is a problem with the coil ID; <i>"The prescribed RF coil and detected RF coil do not match."</i>	Check online instructions by accessing the service browser at Diagnostics; Peripherals; SRI Functional Tests; Coil ID	Start the diagnostic by selecting SRI Coil ID and Run . The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete and the ID is still not displayed, replace the cable.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

3-6 Coil Element Information

Mode	Rcv 1	Rcv 2	Rcv 3	Rcv 4	Rcv 5	Rcv 6	Rcv 7	Rcv 8	PIN 1	PIN 2	PIN 3	PIN 4	PIN 5	PIN 6	PIN 7	PIN 8
8CARDIAC	ON	ON	ON	ON	ON	ON	ON	ON	-5	-5	-5	-5	-5	-5	-5	-5
8CARDIAC POST					ON	ON	ON	ON	+7	+7	+7	+7	-5	-5	-5	-5
8CARDIAC ANT	ON	ON	ON	ON					-5	-5	-5	-5	N/A	N/A	N/A	N/A

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electrical shock may result if the coil is attached to the system during cleaning or when it is wet.



Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 –REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Anterior Cable Replacement

Note: The wrist strap **MUST** be worn all times when handling Coils.

Step 1 – Remove coil cover.

Step 2 – Remove the two screws that hold down the cable assembly.

Step 3 – Disconnect SMB plug and the DC Plug on the cable end from the interface board.

Step 4 – Remove the old cable assembly.

Step 5 – Fix a new cable assembly on the anterior section of the coil and reattach the cable lock with two screws. Connect all four SMB plugs and the DC Plug to the corresponding receptacles on the interface board (see *Figure 12*).

Step 6 – Close the coil cover.

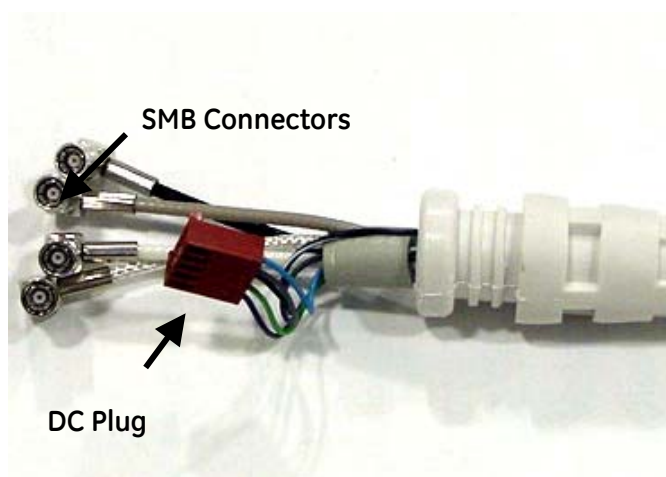


Figure 10: Anterior cable assembly.

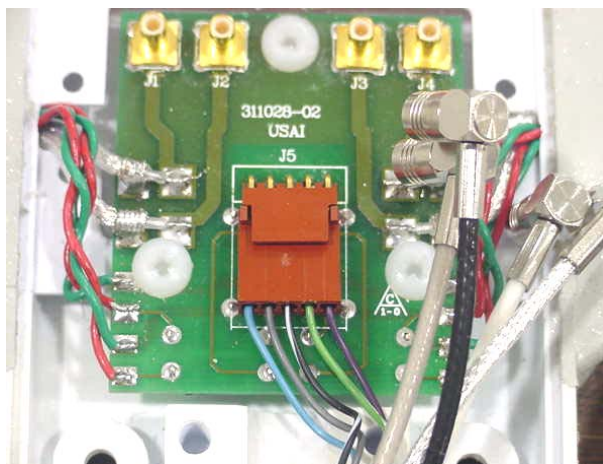


Figure 11: Anterior DC plug assembly.

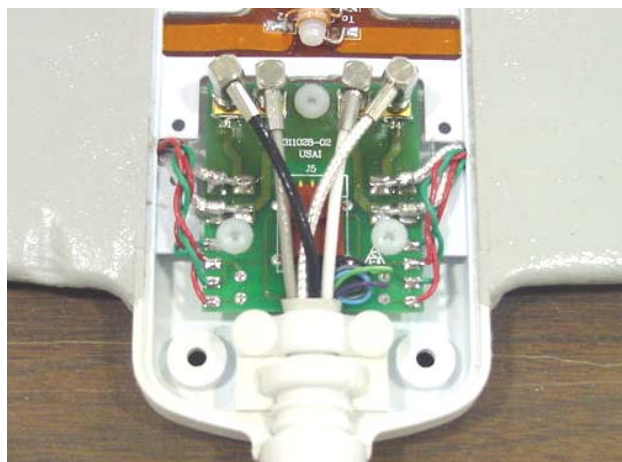


Figure 12: Anterior SMB connector assembly.

5-2 Posterior Cable Replacement

Note: The wrist strap **MUST** be worn all times when handling Coils.

Step 1 – Remove coil cover.

Step 2 – Remove the two screws that hold down the cable assembly.

Step 3 – Disconnect SMB plug and the DC Plug on the cable end from the interface board.

Step 4 – Remove the old cable assembly.

Step 5 – Fix a new cable assembly on the posterior section of the coil and reattach the cable lock with two screws. Connect all four SMB connectors and the DC Plug to the corresponding receptacles on the interface board (see *Figure 15*).

Step 6 – Close the coil cover.

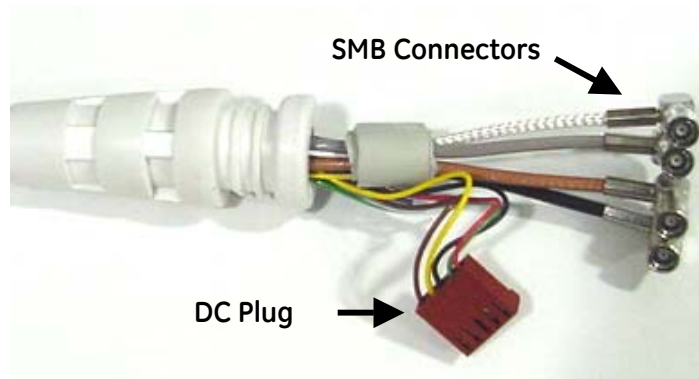


Figure 13: Posterior cable assembly.

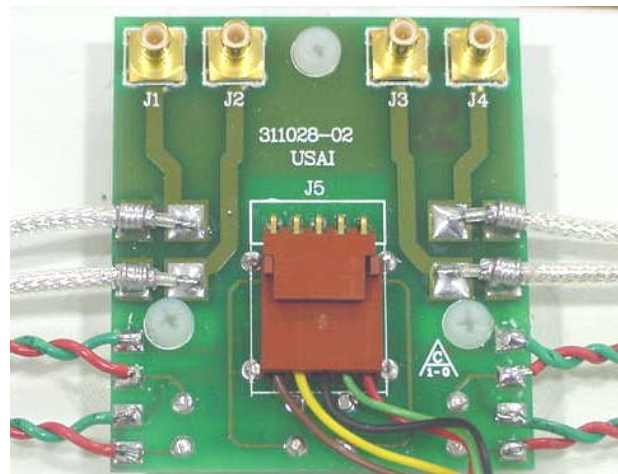


Figure 14: Posterior DC plug assembly.

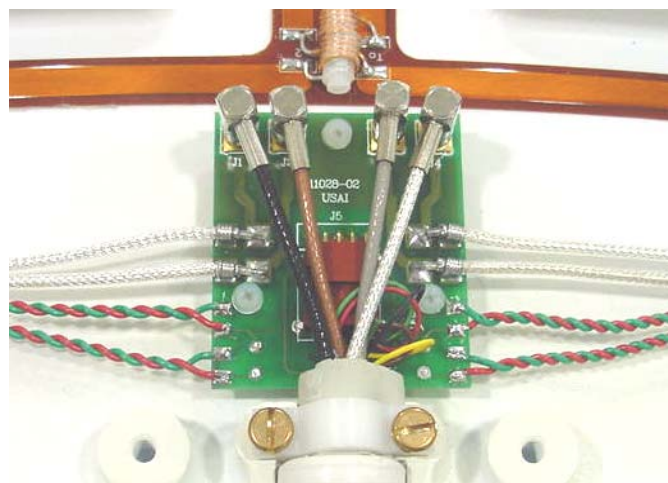


Figure 15: Posterior SMB connector assembly.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #	USAI Part #
Coil	2304255-2	U1-165148
Cable Assembly	2304255-7	U1-155157
Phantom Kit (Phantom, Rings and Positioners)	2304255-3	U1-155158
TL Unified Phantom	5343347	N/A

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #	USAI Part #
Patient Comfort Pad Kit	2304255-6	N/A
Patient Comfort Pad	N/A	U1-155164
Patient Headrest Pad	N/A	U1-155165
Patient Liner Pad	N/A	U1-155166
Velcro Straps (Set of 2)	N/A	U1-155198

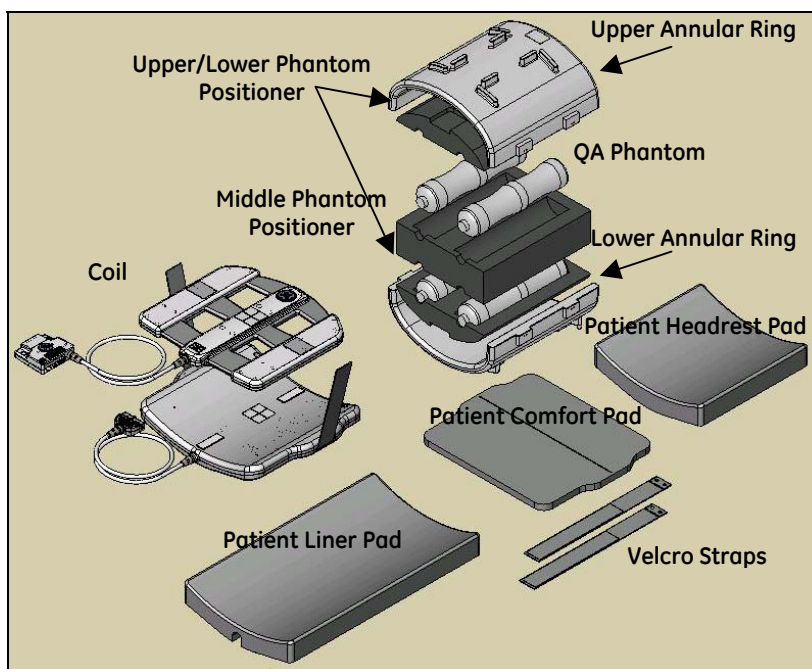


Figure 16: Parts of the coil.

SECTION 7 – APPENDIX

7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks (see section 3).

Date	Configuration	Slice Location	Corresponding Images in Series	Signal Mean	Noise Std. Dev.	SNR

Comments	

Date	Configuration	Slice Location	Corresponding Images in Series	Signal Mean	Noise Std. Dev.	SNR

Comments	

The diagram illustrates a 4-channel audio amplifier system, divided into two identical sections labeled 'Anterior' and 'Posterior'. Each section contains a pre-amplifier stage (P1), a power amplifier stage (P2), and a speaker output stage. The pre-amplifier stage uses a 2N3904 transistor and a 2N3906 transistor. The power amplifier stage uses a 2N3904 transistor and a 2N3906 transistor. The speaker output stage uses a 2N3904 transistor and a 2N3906 transistor. The system is connected to a 'Connector board' and a 'System Connector'.

GE Healthcare Coils
USA INSTRUMENTS

1515 Danner Drive
Aurora, Ohio 44202
USA

www.gehealthcare.com



Imagination at work